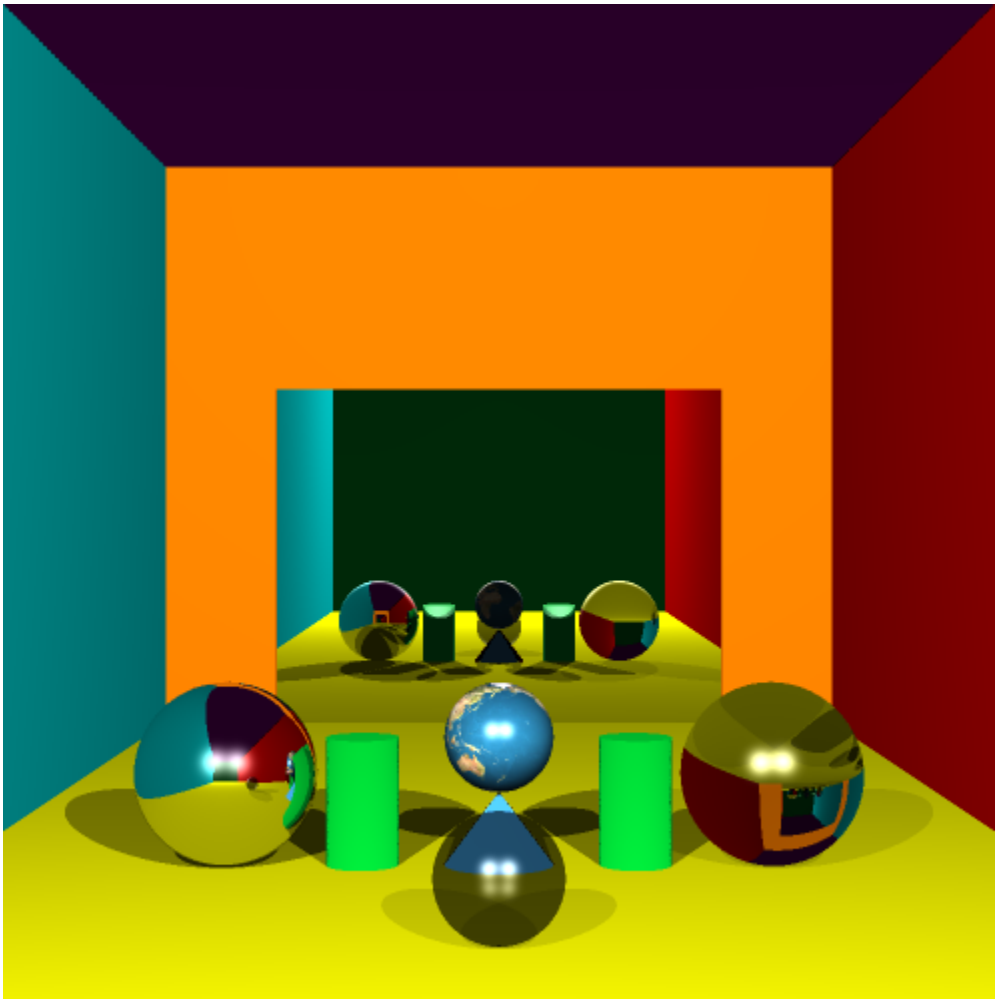
Description

My raytracer includes a multicolored box encapsulating 4 spheres, 2 cylinders, a cone and a mirror (reflective plane). The list of spheres include a reflective, a refractive, a transparent and a textured sphere. There are also 2 separate light sources which generate multiple specular highlights and shadows from objects, along with lighter shadows for transparent and refractive objects.

The project didn't really have any failures as in things I tried to implement but didn't work due to looming deadlines for other projects restricting my time and dissuading me from trying some of the more difficult features.



(with anti-aliasing at 4 samples per pixel)

Extra Features

Cone: Module similar to sphere with an intersect function and normal function.

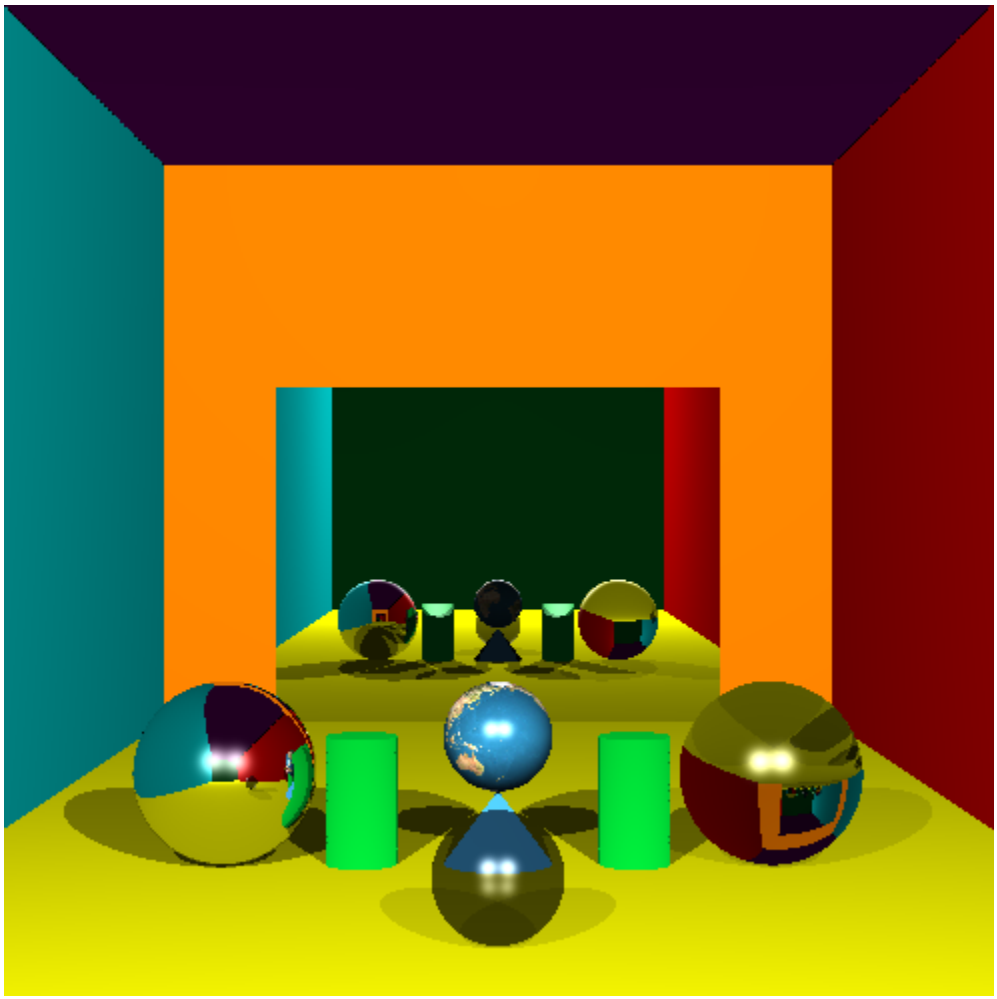
Cylinder: Module similar to sphere with an intersect function and normal function.

Refraction through sphere: changes ray vector direction based on refractive index of object and creates secondary ray to get refracted image.

Super sample anti-aliasing: Splits each pixel into 4 and averages the color values to get the total pixel color.

Textured non-planar object: Earth texture applied to a sphere via alpha and delta angles.

Multiple light sources: A light source in each top corner behind the camera (so 2 total) that create 2 shadows from each object which are darker when they converge.



(without anti-aliasing)

Time estimate ~ 10 seconds to generate an image with anti-aliasing at 4 samples per pixel.

Build process

The build process is the same as the labs. Copy the files into vscodium, transfer the .bmp file into the build folder that is created by cmake, then set the RayTracer.cpp file as the build/launch target to run using cmake.

References

Earth image from https://planetpixelemporium.com/earth.html by James Hastings-Trew.

AI Declaration

The use of AI in the project involved help with debugging and small sections of code at 1 to 3 line snippets from both chatgpt and gemini.

Declaration

I declare that this assignment submission represents my own work (except for allowed material provided in the course), and that ideas or extracts from other sources are properly acknowledged in the report. I have not allowed anyone to copy my work with the intention of passing it off as their own work.

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